WHAT HIDES MUSICAL GEOMETRY IN AUDIO REPRESENTATIONS?

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ABSTRACT

Learned representations are often evaluated by how well their attributes can be decoded. However, linear decodability does not imply that a representation's ambient geometry reflects the relationships among those attributes. Music makes this distinction directly testable because relationships between musical classes, such as keys and pitches, are explicitly defined. We examine this gap through linear decoding, class-centroid geometry, and linear-readout geometry. A log-CQT decodes key better than MERT-v1-330M (.492 vs. .456), yet shows no circle-of-fifths structure in its class centroids while the same structure is strongly expressed in its readout weights. We further find two distinct sources of this mismatch. First, increasing the number of examples per centroid changes fifths geometry by a factor of 13.4, compared with only 1.7 for a thousandfold increase in pooling duration. Second, averaging fails to recover octave equivalence, whereas a rank-4 projection fitted only on GiantSteps keys exposes octave structure in individual NSynth recordings using less than 0.3% of the variance. These results show that decodability, ambient geometry, and readout geometry provide distinct views of musical information in audio representations.

Index Terms— music representation learning, probing, representation geometry, key detection, octave equivalence

1. INTRODUCTION

Self-supervised music audio encoders support linear probes [1] for key, pitch, genre, and instrument [2, 3], and this is usually taken to mean that these attributes are *organized* in the representation. Music provides a useful setting for testing this interpretation because music theory specifies not only which classes exist, but also how they are related. Keys a perfect fifth apart share six of seven scale degrees and are adjacent on the circle of fifths, while pitches an octave apart belong to the same pitch class.

We compare what can be decoded from a representation, how musical classes are arranged by distances in the original representation space, and how those classes are arranged in the weights of a linear classifier. Across these measurements, linear accessibility does not necessarily correspond to structure in the original representation geometry. Our results trace this mismatch to two distinct sources. For clip-level key geometry, the dominant factor is the number of examples used to estimate each class centroid rather than the duration of each example. For note-level octave equivalence, increasing the number of examples does not recover the structure, whereas a learned projection exposes it under a different metric. Together, these results show that the absence of structure in the original geometry does not by itself imply the absence of that structure from the representation.

Code, pre-registrations and all run artefacts: `https://github.com/Junst/topo-music`. Every threshold and kill criterion reported here was registered before the corresponding run.

2. RELATED WORK

Linear probes are the standard instrument for asking what a representation encodes [1], and music audio encoders are routinely evaluated this way, with MERT [2] and MARBLE [3] reporting probe accuracy for key, pitch, genre and instrument. Hewitt and Liang [4] show that a probe's success has to be read against a control task that removes the structure of interest, and we adopt that design for representation geometry rather than for probe complexity.

Music cognition has established well-defined relational structures for both key and pitch. Perceived key relationships follow the circle of fifths and Krumhansl–Kessler profiles [5, 6], while pitch perception separates pitch height from chroma [7]. Chroma features [8, 9] encode octave equivalence by construction and therefore provide a positive control for these relationships. Tonality-aware objectives such as STONE [10] explicitly impose key equivariance during training. Pilataki et al. [11] report a closely related observation for a different encoder family, clustering NSynth embeddings from a JEPA transcription model and finding timbre and pitch height well separated while pitch class is not.

3. METHOD

Representation decomposition. We write an item of class y as

$$z = s_y + n_y + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid y] = 0, \qquad (1)$$

where s_y represents the class-dependent signal whose relations encode the structure of interest, such as key relationships or octave equivalence, n_y captures other variation that depends on the class, and ε captures variation whose conditional mean is zero within each class. Averaging examples from the same class reduces ε but not n_y, whereas a change of metric can alter how s_y and n_y contribute to the measured geometry.

Representations. MERT-v1-330M [2] layers 4 to 24, EnCodec 32 kHz [12], as used by MusicGen [13], and a log-CQT [8, 9] as a spectral baseline. A CQT chromagram [9], which folds frequencies across octaves into 12 pitch classes, provides a positive control for octave-invariant structure. Frame-level representations are averaged over time to obtain one vector per audio clip.

Key-centroid geometry. On GiantSteps [14], with 7035 clips covering all 24 keys, we average the representation vectors of tracks sharing the same key to obtain 24 class centroids and rank-correlate their pairwise Euclidean distances against three reference structures: circle-of-fifths distance, the Krumhansl–Kessler key-profile distance, and semitone distance. Significance uses a 2000-draw permutation null over key labels, and a partial Spearman additionally removes mel-spectral centroid distance. Semitone distance is included as a control to distinguish circle-of-fifths structure from proximity in chromatic pitch space.

Readout geometry. The same probe, a multinomial logistic regression under 5-fold cross-validation, yields 24 class weight vectors, and we apply the identical measurement to their pairwise cosine distances.

Octave equivalence. On NSynth [15] we use real recordings only, since pitch shifting by signal processing would build the answer into the stimulus. For each instrument we fix an anchor set of pitches supporting every transposition up to 25 semitones, so the anchor set is constant across transpositions, and measure the mean cosine similarity between a note and its transposition by k semitones. Three statistics are reported together, as registered in advance: a local octave contrast, comparing the similarity at 12 semitones against the mean of 11 and 13; a strict contrast, comparing it against the largest similarity anywhere from 8 to 11 semitones; and the coefficient on a twelve-semitone periodic term in a fitted curve. Confidence intervals are bootstrapped over *instruments*. We require the three to agree in sign, since a positive periodic coefficient with no local peak at 12 semitones is a curve-fitting artefact.

Subspaces. Each subspace is an orthonormal basis of the top d shrinkage-LDA discriminant directions for a stated target, with d at most 11, since any linear map to a twelve-class target has rank at most 11. We report the fraction of total variance the subspace holds and, as a control, matched random orthonormal subspaces, because projection changes the metric whatever the directions are.

Table 1. Key information, ambient geometry and readout geometry come apart. Same 7035 GiantSteps clips, same 24 keys. *acc* is 24-way key accuracy (chance .042); ρ_5^{cent} is the circle-of-fifths correlation among the 24 key centroids, i.e. the ambient metric; ρ_5^{W} is the same correlation among the probe's 24 class weight vectors, i.e. what a linear readout reaches. The controls fix the scale for reading the two geometry columns.

representation	acc	ρ_5^{cent}	ρ_5^{W}
Controls			
analytic fifths (metric ctrl)	0.274	$+0.977$	$+0.986$
random features (no info)	0.055	$+0.009$	$+0.033$
orthogonal key codes (no tonal rel.)	0.651	-0.008	-0.005
Spectral front ends			
log-CQT, 84 bin, dB	0.492	-0.035	$+0.560$
log-CQT, 84 bin, per-frame norm	0.486	-0.117	$+0.671$
octave-summed, 12 bin, dB	0.407	-0.106	$+0.305$
chromagram (fold + norm)	0.451	$+0.475$	$+0.692$
Neural codec			
EnCodec 32 kHz	0.404	$+0.177$	$+0.648$
MERT-v1-330M			
layer 4	0.451	$+0.445$	$+0.412$
layer 12	0.456	$+0.468$	$+0.428$
layer 16	0.440	$+0.395$	$+0.409$
layer 24	0.418	$+0.294$	$+0.484$

4. RESULTS

4.1. Information, metric and readout dissociate

Table 1 measures all three on the same clips and keys, and the control block sets the scale for reading it. An analytic circle of fifths scores close to 1 in every column, so a null elsewhere is a null of the representation and not of the method, while random features show nothing anywhere. Orthogonal per-key codes are highly decodable with tonal relations absent by construction. They decode key at .651, exceeding every real representation, and yet show no fifths geometry in either their centroids, at -0.008, or their readout weights, at -0.005. This follows the logic of control tasks [4] without relying on matched accuracy, and this rules out an alternative explanation: that any sufficiently accurate 24-way classifier acquires fifths-structured weights because neighboring keys are intrinsically confusable.

The dissociation that follows is unambiguous, in that the log-CQT has the highest key accuracy of any real arm and no fifths structure in its centroids, at -0.035 with p of .52, while its probe weights carry that structure strongly, at 0.560 with a z of 9.1. Tonal relations are therefore linearly accessible in a log-CQT without being expressed in its ambient centroid geometry.

For MERT, the main change is in ambient geometry rather than linear accessibility. Its readout geometry, between 0.41 and 0.48, is *lower* than that of the log-CQT, EnCodec or the chromagram, while its centroid geometry is far higher. Across depth, tonal relational geometry weakens, from 0.445 at layer 4 to 0.294 at layer 24, while major and minor become more

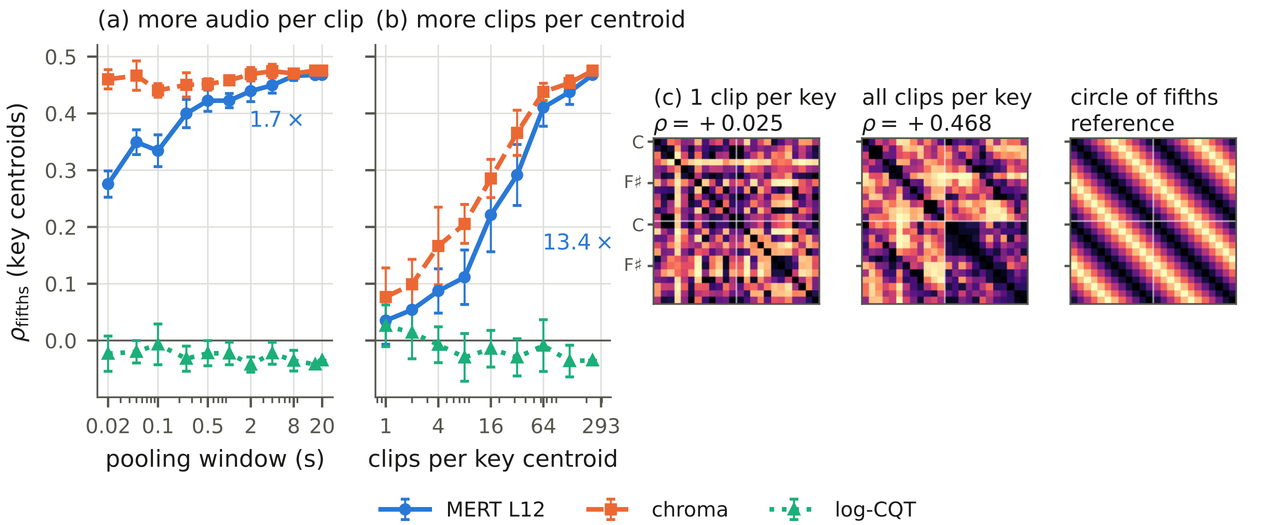

Fig. 1. Clip-level key geometry is set by centroid sample size, not by temporal context. Panels (a) and (b) plot the same quantity on the same vertical axis. (a) A thousandfold longer pooling window moves MERT L12 by a factor of 1.7 and moves the chromagram by 3%, and a single 13 ms frame already carries 59% of the 20 s value. (b) Increasing the number of clips averaged into each key centroid moves it by a factor of 13.4, and in the single-example centroid regime the geometry is absent. Error bars are one standard deviation over 5 draws in (a) and 10 in (b). (c) The same rise seen as geometry: pairwise distances between the 24 MERT L12 key centroids, ordered round the circle of fifths, majors above the white rule and minors below, beside the reference they are correlated against. Each panel is rank-transformed over its own entries, which is what a Spearman correlation sees. At one clip per key there is no visible band, and with every clip a band aligned with the reference appears in each quadrant. The single-clip panel is the median of ten draws.

separable, from 0.216 to 0.328. We report this as a correlation rather than a mechanism, since key accuracy does not improve along the way, and the mode effect is not significant in the chromagram control, so the growing mode axis is not pitch-class based.

4.2. Sample-limited masking

The clip-level result invites a temporal reading, in which keys are properties of seconds of music, but that reading does not survive the measurement. Fig. 1(a) sweeps the pooling window from one frame to the full 20 s clip, drawing one window per clip at every length so the number of clips behind each centroid is held constant. A single 13 ms MERT frame already yields a correlation of 0.276, which is 59% of the 20 s value, and the chromagram is flat to within 3%, with no knee and no threshold anywhere in the sweep.

The clip-level measurement averages twice, and only one of those averages was being varied. Fig. 1(b) holds the window at the full clip and sweeps the number of clips per centroid, giving 0.035 at one clip and 0.468 at all of them for MERT L12, a factor of 13.4 against 1.7 for a thousandfold longer window. In the single-example centroid regime, fifths geometry is indistinguishable from zero for every arm. Fig. 1(c) shows what that correlation is a summary of, in the centroid distance matrices themselves.

This is Eq. (1) with ε dominant, because what differs between two GiantSteps tracks in the same key, namely production, instrumentation and subgenre, is largely unrelated to the key and cancels in the mean. Centroid sample size is therefore the dominant source of the apparent clip- level effect, and a longer window accounts for a small fraction of it.

4.3. Metric masking

If low between-item signal-to-noise were the whole story, the same averaging should expose octave equivalence on NSynth, and we pre-registered that prediction in advance of the run that refutes it. Across a sweep over notes per pitch centroid, every arm saturates by four notes at approximately zero, MERT L24 remains negative, and the chromagram needs no averaging at all, reaching 0.566 from single notes. Register and spectral envelope are functions of pitch, so they survive into every pitch centroid, and averaging over instruments cancels the instrument but never the octave, which is the behaviour Eq. (1) attributes to n_y.

A projection can nonetheless expose the target relation despite n_y, and the evidence here is cross-dataset and cross-task. We fit LDA directions on GiantSteps track tonics only, with no notes, no NSynth and no octave information, and apply them unchanged to individual NSynth recordings with no averaging. At rank 4 the strict octave contrast moves from

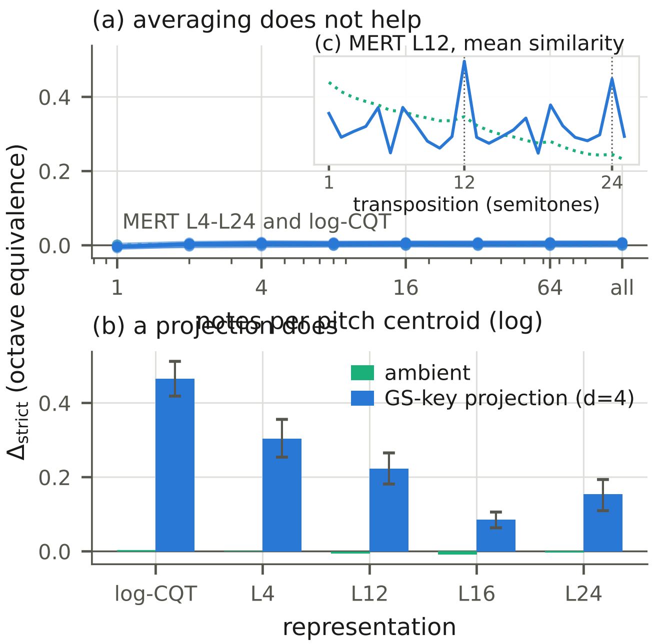

Fig. 2. Two operations on NSynth notes, with one outcome each. Panels (a) and (b) share a vertical axis, since rescaling (a) would invent structure that is not there. (a) Averaging notes of the same pitch into a pitch centroid does not produce octave equivalence at any sample size, with every MERT layer and the log-CQT flat at zero. The chromagram is off scale throughout, moving only from 0.57 to 0.62, since it has octave equivalence already in single notes. (b) A rank-4 projection fitted only on GiantSteps track tonics, applied unchanged to individual recordings, with instrument-bootstrap 95% intervals. (c) The curve (b) summarises, for MERT layer 12 and in the colours of (b), standardised within each condition.

-0.005 to 0.223 for MERT L12, from -0.001 to 0.304 for layer 4, and from 0.004 to 0.466 for the log-CQT, with the other two statistics agreeing in sign and bootstrap intervals excluding zero. Fig. 2(c) shows the curve those contrasts are drawn from, where the projected similarities reach maxima at exactly twelve and twenty-four semitones against an ambient baseline that decays smoothly. Matched random subspaces stay between -0.019 and 0.007, and the subspace uses between 0.11 and 0.24% of MERT's variance and 1.85% of the log-CQT's. The same absence has been read as a property of the pretraining objective, with a change of masking proposed as the remedy [11], whereas the transfer here exposes the relation in a fixed representation without retraining.

Key supervision does not guarantee pitch-height invariance, because GiantSteps key labels may correlate with subgenre, tempo, or production. The cross-dataset transfer nevertheless rules out an explanation based only on GiantSteps-specific nuisance correlations. Nor is it a mechanistic claim, since whether the projection suppresses class- coupled variation or amplifies the tonal component is not identifiable from this measurement.

A registered prediction that failed. We predicted that removing the subspace from the representation would at least halve clip-level fifths geometry. Against a baseline recomputed on the same held-out clips it does not move it, going from 0.127 to 0.119 for MERT L12 and from 0.118 to 0.114 for layer 4, and only the twelve-dimensional chromagram collapses, which is a dimensionality effect. A subspace holding well under 1% of the variance is therefore sufficient to expose the relation, but the relation is not localised exclusively to the one we recover. How widely it is distributed would need repeated recovery after sequential deflation, which we do not attempt here.

5. DISCUSSION

Information availability, metric visibility and readout accessibility are separable, and separating them changes what a probe result licenses: high accuracy says the structure is linearly recoverable, and nothing about whether it governs the distances a similarity or retrieval system consumes.

The two masking modes suggest different remedies. Sample-limited masking is a property of the evaluation, and the fix is more examples per class, not longer audio. Reporting the number of items behind each centroid should be standard, since the same encoder scores 0.035 or 0.468 depending only on that. Metric masking persists under additional averaging and instead requires changing how distances are measured, for example through a learned transform.

Nothing in Eq. (1) is specific to music, since semantic relations in vision may be linearly available while the geometry is dominated by pose or background, and phonetic structure in speech may be masked by speaker and channel.

Limitations. GiantSteps is entirely electronic dance music, where key correlates with subgenre. The cross-dataset transfer constrains that confound but does not remove it, and a key-balanced, genre-diverse corpus would settle it. NSynth is monophonic and drawn from isolated instruments, so its nuisance structure is unusually clean. Our subspaces are linear by design, and a nonlinear transform might expose more at the cost of interpretability. The mode effect is reported but not explained.

6. CONCLUSION

Musical structure can be present in a representation, linearly accessible and still invisible in its geometry, for two reasons calling for two different operations. Averaging over examples accounts for most of the apparent clip-level key geometry and distinguishes sample-limited estimates from stable metric effects, while a rank-4 projection fitted on another dataset and task exposes octave equivalence that no amount of averaging reaches, testing whether the relation remains recoverable outside the ambient metric.

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